fragmentation, sporangiospores, conidia etc., Meiosis is zygotic. Example : *Peziza*

Recently, with the advent of molecular methods myxomycetes and oomycetes were reclassified and treated under chromista.

The salient features of some of the classes – Oomycetes, Zygomycetes, Ascomycetes, Basidiomycetes and Form class Deuteromycetes are discussed below.

**Oomycetes**

Coenocytic mycelium is present. The cell wall is made up of Glucan and Cellulose. Zoospore with one whiplash and one tinsel flagellum is present. Sexual reproduction is oogamous. Example: *Albugo.*

**Zygomycetes**

- Most of the species are saprophytic and live on decaying plant and animal matter in the soil. Some lead parasitic life (Example: *Entomophthora* on housefly).
- Bread mold fungi (Example: *Mucor, Rhizopus*) and coprophilous fungi (Fungi growing on dung Example: *Pilobolus*) belong to this group (Figure 1.20).

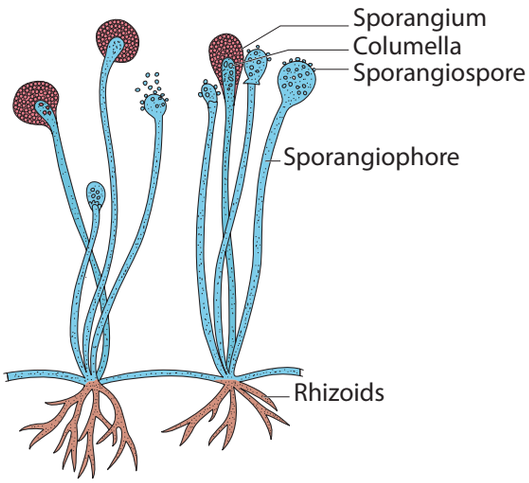
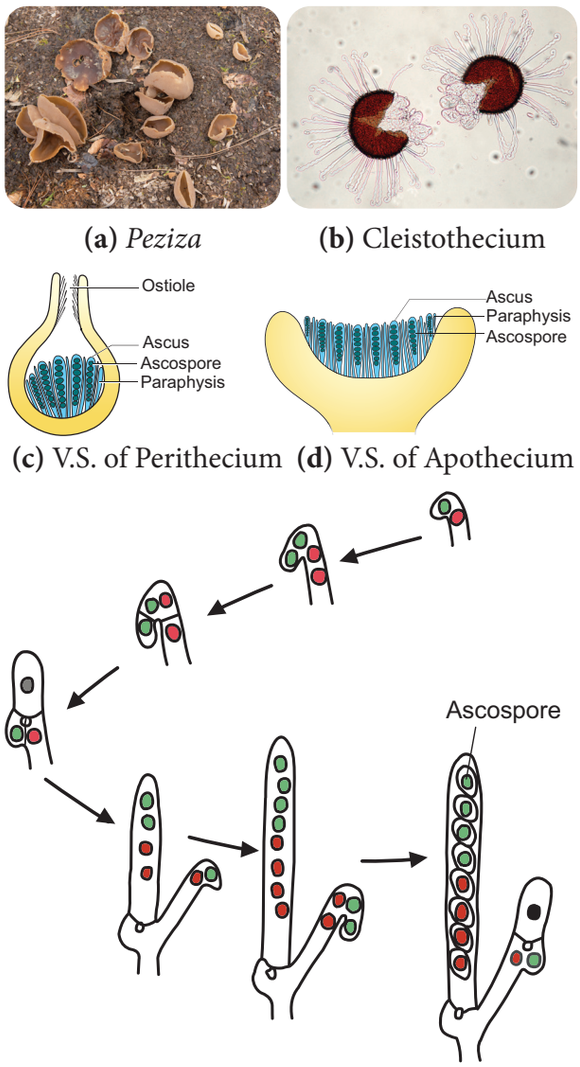



**Figure 1.20:** Zygomycetes – *Rhizopus*

- The mycelium is branched and coenocytic.
- Asexual reproduction by means of spores produced in sporangia.
- Sexual reproduction is by the fusion of the gametangia which results in thick walled zygospore. It remains dormant for long periods. The zygospore undergoes meiosis and produce spores.

**Ascomycetes**

- Ascomycetes include a wide range of fungi such as yeasts, powdery mildews, cup fungi, morels and so on (Figure 1.21).




**(a)** *Peziza* **(b)** Cleistothecium

**(c)** V.S. of Perithecium **(d)** V.S. of Apothecium

**(e)** Steps involved in the development of Ascus

**Figure 1.21:** Structure and reproduction in Ascomycetes

- Although majority of the species live in terrestrial environment, some live in aquatic environments both fresh water and marine.
- The mycelium is well developed, branched with simple septum.
- Majority of them are saprophytes but few parasites are also known (Powdery mildew – *Erysiphe*).



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